Mathematical Foundations

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Instructions

This exam has 6 sections. Provide the answers to each section on a new page, labeling the part of the question for each answer.

I PROBABILITY AND MATHEMATICAL LOGIC

1) If $P(A \cap B) = 0$, which of these are false? (there may be more than one)
   a) $P(A|B) = P(A)$ Is FALSE
   b) $P(A|B^c) = \frac{P(A)}{P(B^c)}$ IS TRUE
   c) $P(A) + P(B) = P(A \cup B)$ IS TRUE
2) Why is it that:
   a) If $A \subset B$ then $P(B|A) = 1$, because If A happens then B is sure to have happened. $P(A \cap B) = P(A)$ so $P(B|A) = P(A)/P(A) = 1$
   b) If $A \subset B$ then $P(A|B) = \frac{P(A)}{P(B)}$. $P(A \cap B) = P(A)$ so $P(A|B) = P(A)/P(B)$
3) Roll three six-sided dice. Find the probability that there are at least two sixes given that there is at least one six. (hint: use the equation in question 2.b).

   Call A the event where there is at least one six. Call B the event where there are at least two six. $B \subset A$, so $P(B|A) = \frac{P(B)}{P(A)}$ So P(at least two six— at least one six)=(3*1/6*1/6*5/6+1/6*1/6*1/6)/(1-5/6*5/6*5/6)
4) TRUE or FALSE, if false, explain in one sentence.
   a) $\bar{X} = \mu$ FALSE $E(\bar{X}) = \mu$
   b) If X and Y are independent, $Var(X - Y) = Var(X) + Var(Y)$ TRUE
   c) $\bar{X} \sim N(0, 1)$ FALSE $\bar{X} \sim N(\mu, \sigma^2/n)$
   d) $Var(\mu) = 0$ TRUE
   e) $Var(\bar{X}) = \frac{\sigma^2}{n}$ TRUE

II Basketball!

Since 2010, the Women's National Basketball Association has a best of five series for the National Championship. This means that the series can end only when one team wins 3 games. We are interested in the number of games that are played for one team to be named Champion.

1) If both teams are equally likely to win, what is the probability that the series lasts to exactly 4 games? 3/8
2) What is the probability that the series lasts less than 5 games? 1-3/8
3) What is the expected number of games played? 1/4*3+3/8*4+3/8*5=4.125

III Triangle Distribution

Suppose that

$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 4x & \text{for } 0 \le x < 1/2 \\ 4 - 4x & \text{for } 1/2 \le x \le 1 \\ 0 & \text{for } 1 < x \end{cases}$$

1) Verify that f(x) is a pdf. Draw f(x). It is.
2) What is the probability that $X > .7$? $\int_{.7}^{1} 4 - 4x dx = 4x - 2x^2|_{.7}^{1} = 2 - 1.82 = .18$
3) Calculate the median of X. .5

IV Surveying Sensitive Information.

It is difficult to conduct surveys on sensitive issues because many people will refuse to give answers that might embarrass them. "Randomized response" is a method designed to ensure anonymity when asking about topics such as drug use. To ask a sample of college students whether they have used an illegal drug, have each student toss a coin in private. If the coin lands heads and they have not used illegal drugs, they are to answer "no." Otherwise, they are instructed to answer "yes." Only the student knows whether the answer reflects the truth or the coin toss.

1) Suppose 25% of the students have used illegal drugs. What is the probability of a 'no'? p("yes"—"no drugs") * p(no drugs)+ p("yes"—"drugs")*p(drugs)= .5*(.75)+ .5*.25= .5
2) Now suppose that you conduct the survey and find that 35% of the students give a 'no' answer. What would be your estimate of the percent of students who have used illegal drugs? %70

V Calculating Expectations

1) Suppose that the sample space S contains three elements $\{1,2,3\}$, with probabilities .5, .2, and .3 respectively. Define a random variable X as follows:

$$X(s) = s^2 - 4 \text{ for } s \in S$$

   a) What is E[X]? .5*(1-4)+.2*(4-4)+.3*(5)=0
   b) What is Var(X)?

2) In class we examined a random variable $Z \sim N(0,1)$
   a) What is E[Z^2]? 1
   b) What is E[Z^3]? 0 (Take the derivative of the MGF)

3) Suppose you enter a casino, and they offer you one of two bets:

   A Roll a six-sided die 5 times, with sides printed $\{1,\ 2,\ 3,\ 4,\ 5,\ 6\}$, take the average of your rolls and receive the number of dollars equal to the average, translated by the following equation:

$$e^{\bar{x}-3.5}$$

   B Roll a six-sided die 5 times, where each of the sides of the die has $e^{x-3.5}, \quad x \in \{1,\ 2,\ 3,\ 4,\ 5,\ 6\}$ written on it. i.e. $\{e^{-2.5},\ e^{-1.5},\ e^{-.5},\ e^{.5},\ e^{1.5},\ e^{2.5}\}$, and receive the average of the number of dollars on its face.

   a) Which of these bets would you take, **A** or **B**? Why? I prefer B because of Jensen's inequality.
   b) What if the equation were $\sqrt{x}$? $log(x)$? Then I would prefer these A.

VI Limit Theorems

1) Suppose $\bar{X}$ is the sample mean of 100 observations from a population with mean μ and variance σ^2. Find 90% confidence limits for $\bar{X}$.
   a) Using Chebychev.
   b) Using Central Limit Theorem.
   c) Compare the two limits.

2) Suppose $\bar{X}_1$ and $\bar{X}_2$ are two independent surveys of size n from the same population. Assume we further know that for the individual responses, $Var(X_i) = \sigma^2$.
   a) What is the distribution of $\bar{X}_1 - \bar{X}_2$? (hint: see problem 4 from section I)
   b) Suppose you had to pick a sample size such that:

$$P(|\bar{X}_1 - \bar{X}_2| < \sigma/5) \approx .99$$

   Use the fact that $\Phi^{-1}(.995) = 2.576$ to decide how big of a sample you should use.